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Partitioning of Oxygen Uptake between Cutaneous and Branchial Surfaces in Larval and Young Juvenile Chinook Salmon *Oncorhynchus tshawytscha*

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Abstract

Rates of O₂ uptake across the skin and across the gills of young chinook salmon Oncorhynchus tshawytscha weighing between 92 mg (shortly after hatch) and 1.1 g (well into the fry stage) were determined at 12°C with a two-chamber, closed-system respirometer. Cutaneous uptake accounted for most of the O₂ consumed by newly hatched alevins (up to 84%). However, the branchial contribution increased rapidly during larval development, so that by the time alevins weighed about 281 mg (halfway through the alevin stage), only 50% of the total O₂ consumed was taken up across the skin. This trend continued; by the time the alevins began exogenous feeding (≈ 400 mg), cutaneous uptake had fallen to 40% of total O₂ consumption. The relative importance of the skin declined more slowly during the fry stage and appeared to stabilize at about 30% of total uptake. About half of the increase in the proportion of O₂ taken up across the gills was due to a relative increase in branchial surface area. The other half was due to an increase in the efficiency of branchial uptake as the fish matured. In newly hatched larvae, area-specific O₂ uptake across the gills was only slightly higher than that across the skin (10.1 vs. 7.5 μg · h⁻¹ · cm⁻²), whereas in a 1-g fry roughly six times as much O₂ per unit area was taken up across the gills as across the skin (60 vs. 9.6 μg · h⁻¹ · cm⁻²).

Introduction

A substantial fraction (up to 35%) of total O₂ uptake ($\dot{M}_{O_2}$) takes place across the skin in juvenile and adult fish (Kirsch and Nonnotte 1977; Steffensen, Lomholt, and Johansen 1981; Toulmond, Dejours, and Truchot 1982; Steffensen and Lomholt 1985; LeMoigne et al. 1986). There are reasons to believe that the skin plays an even more important role during larval de-

velopment. Cutaneous surface area (A_c) is relatively much larger in larvae than in older fish (DeSilva 1974; Oikawa and Itazawa 1985; Rombough and Moroz 1990). The skin, in particular that on the yolk sac and the fins, is well vascularized, and, compared with older fish, diffusion distances appear to be relatively small (Rombough 1988c). Blood flow in the skin of at least some species is in the opposite direction to the normal flow of water past the larvae, allowing the larvae to take advantage of the high exchange efficiency inherent in such a countercurrent arrangement (Liem 1981). Alternate sites for gas exchange are severely limited in larvae. The gills, which are the major site of gas exchange in adult fish, are not even present in the larvae of some species (e.g., herring *Clupea harengus*; DeSilva 1974). In other species, gills are present but studies of their morphology suggest that they are relatively inefficient gas exchangers. Diffusion distances tend to be greater than in the adult gill (Morgan 1974), lamellar surface area tends to be relatively small (DeSilva 1974; Rombough and Moroz 1990), and ventilatory movements may be poorly coordinated (Holeton 1971; Peterson 1975).

Although the available evidence would suggest that the skin plays a major role in larval gas exchange, no one to our knowledge has actually measured cutaneous uptake in teleost larvae. The aim of this study, thus, was twofold: (1) to determine the extent of cutaneous O_2 uptake ($\dot{M}_{O_2c}$) in chinook *Oncorhynchus tshawytscha* alevins (larvae) and (2) to monitor changes in the partitioning of $\dot{M}_{O_2}$ between the skin and the gills as the alevins mature. Chinook alevins were chosen for this study because their comparatively large size makes them amenable to partitioning studies and because we already had information on the surface areas of the various cutaneous and branchial structures involved in gas exchange during the course of larval and early juvenile development.

Material and Methods

Gametes of chinook salmon were obtained from fish returning to spawn in the Quinsam River, Vancouver Island, British Columbia in November 1987 and November 1988. Fertilized eggs were incubated in virtual darkness in dechlorinated City of Brandon tap water (total hardness $\approx 240 \text{ mg} \cdot \text{L}^{-1}$ as CaCO_3). Embryos and alevins were raised initially at low temperatures (mean temperatures $\pm$ SD of $1.5^\circ\text{C} \pm 0.3^\circ\text{C}$ and $2.1^\circ\text{C} \pm 0.3^\circ\text{C}$, respectively, in 1987/1988 and 1988/1989). Beginning just before hatch, groups of fish were transferred at intervals to higher temperatures ($10.9^\circ\text{C} \pm 1.4^\circ\text{C}$ and $8.5^\circ\text{C} \pm 1.6^\circ\text{C}$, respectively, in 1987/1988 and 1988/1989) to speed up the rate

of development. This procedure allowed replicate testing of fish at equivalent stages of development. After "swimup," fish were exposed to a 12L:12D cycle and fed a diet of commercial dry feed and fresh beef liver.

Oxygen uptake across the skin and across the gills of young chinook was measured at intervals between hatch and midway through the fry stage with a closed system respirometer. The respirometer consisted of two identical 15.2-mL glass chambers held together at the frosted ends by pinch clamps (fig. 1). The respirometer was immersed in a 45-L water bath held at a constant temperature of $12.0^{\circ}\text{C} \pm 0.1^{\circ}\text{C}$. Polarographic electrodes (Orbisphere model 2120) were inserted into the ends of each chamber. O-rings and petroleum jelly insured a gastight seal. Magnetic stirrers were used to maintain uniform mixing of the water in the chambers and a constant flow past the electrodes. Signals from the polarographic electrodes were sent to a multirange, two-channel O₂ meter (Cyclobios), and the output was recorded on a flat-bed chart recorder.

A 0.25-mm rubber dental dam was used to separate the chambers. Preliminary tests showed that the diffusion of O₂ across the membrane during a 1-h period was negligible at diffusion gradients of up to 140 Torr. Fish were inserted into an appropriately sized hole in the center of the rubber dam. The membrane was positioned just posterior to the pectoral fins. As a result, the bulk of A_c (>75%) was isolated in the posterior chamber while the water

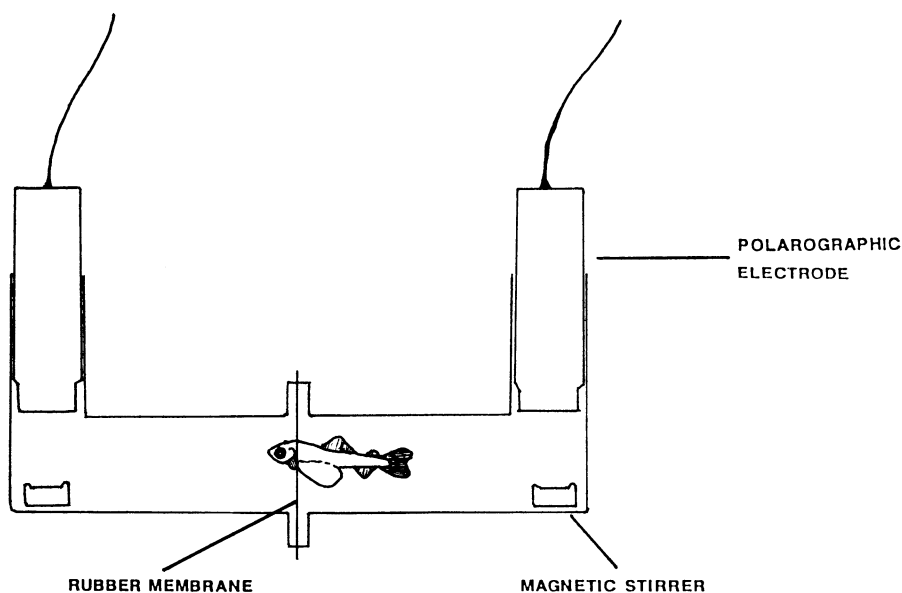


Fig. 1. Schematic diagram of the respirometer used to determine O₂ uptake rates of chinook alevins and fry.

passing over the gills remained in the anterior chamber. If the aperture was of the correct size and the membrane was positioned properly, there was no significant leakage between the two chambers (any leakage was readily apparent in the polarographic record).

Fish were acclimated to the test temperature (12°C) for at least 6 h prior to the start of an experiment. About 30 min prior to a test, the fish were lightly sedated with pH-adjusted (7.0–7.5) tricaine methanesulfonate (MS222) to minimize stress during membrane placement and to reduce activity during the experiment. Levels of MS222 ranged from 8 to 35 mg · L⁻¹, with the exact concentration depending on the size of the fish (larger fish required higher concentrations to maintain the same level of anesthesia). Previous studies showed that $\dot{M}O_2$ was not affected by MS222 concentrations of less than 56 mg · L⁻¹ (P. J. Rombough, unpublished data). Fish were allowed 20 min to recover after membrane placement. Any fish that displayed overt signs of distress during this period were not used.

All tests were conducted under subdued lighting. In most cases, initial O₂ levels in the test chambers were close to 100% of the air-saturated value (ASV). In tests involving larger fry, however, initial O₂ levels were elevated to about 150% of ASV so that O₂ levels in the respirometer would not fall below the critical level (about 35% of ASV at 12°C; Rombough 1988a) before uptake rates could be determined. The respirometer was kept sealed until the O₂ concentration in the chamber in which O₂ was consumed most rapidly fell to about 25%–30% of ASV. On average, this took about 1 h. At the end of the test, the fish was removed from the membrane and allowed to recover in MS222-free water. If the fish did not appear to behave normally after 30 min, the results of that test were discounted. Fish that behaved normally were fixed in 10% neutral buffered formalin for at least 48 h before being dissected into tissue and yolk components. Tissue and yolk (if present) were weighed separately.

Rates of O₂ uptake were estimated from the slopes of the polarographic record according to a procedure similar to that described by Rombough (1988b). As noted previously (Rombough 1988b), O₂ levels in closed respirometer chambers typically decline for a while in a curvilinear fashion with respect to time before stabilizing at a more or less constant rate of O₂ depletion. With our apparatus it usually took about 12 min for uptake rates to stabilize. We therefore took the longest linear portion of the polarographic record, exclusive of the first 12 min, as our estimate of routine metabolic rate. In all cases, this coincided with O₂ levels of between 100% of ASV and 37% of ASV (the critical level).

Tests conducted without fish were used to correct for microbial O₂ consumption. In addition, seven tests using various size fish were conducted

without the rubber dam. By comparing $\dot{M}_{O_2}$'s determined in this manner with those estimated by adding the $\dot{M}_{O_2}$'s in the posterior and anterior chambers for similar-size fish, we were able to check whether our experimental procedure had a significant effect on total O₂ consumption.

Rates of O₂ uptake in the anterior and posterior compartments were transformed to provide estimates of cutaneous and branchial contributions to total O₂ consumption. Total $\dot{M}_{O_2c}$ ($\mu\text{g} \cdot \text{h}^{-1}$) was estimated as

$$\dot{M}_{O_2c} = \dot{M}_{O_2p} / (A_{cp} / A_{ct}), \quad (1)$$

where $\dot{M}_{O_2p}$ is O₂ uptake in the posterior chamber ($\mu\text{g} \cdot \text{h}^{-1}$), A_{cp} is cutaneous surface area in the posterior chamber (mm^2), and A_{ct} is total cutaneous surface area (mm^2). The fraction of total cutaneous surface area in the posterior compartment was estimated as

$$A_{cp} / A_{ct} = 0.7587 - 0.1103 \log m \quad (R^2 = 72.5\%, n = 40), \quad (2)$$

where m is tissue mass (g). Equation (2) is based on morphometric data, summarized in Rombough and Moroz (1990), for siblings of the fish used in this experiment. Branchial uptake ($\dot{M}_{O_2b}$; $\mu\text{g} \cdot \text{h}^{-1}$) was estimated by subtracting $\dot{M}_{O_2c}$ from total (anterior + posterior) uptake:

$$\dot{M}_{O_2b} = \dot{M}_{O_2t} - \dot{M}_{O_2c}, \quad (3)$$

where $\dot{M}_{O_2t}$ total uptake is ($\mu\text{g} \cdot \text{h}^{-1}$).

Area-specific $\dot{M}_{O_2}$'s across the skin and the gills were estimated by dividing the $\dot{M}_{O_2c}$'s and $\dot{M}_{O_2b}$'s, respectively, by the total A_c 's and branchial surface areas (A_b 's) characteristic of that particular size fish. Cutaneous surface area (mm^2) was estimated as

$$A_c = 347.3 + 601.9 m \quad (R^2 = 92.1\%, n = 41), \quad (4)$$

while branchial surface area (mm^2) was estimated as

$$A_b = -4.619 + 923.9 m - 1,175 m^2 + 1,655 m^3 \quad (R^2 = 96.7\%, n = 41). \quad (5)$$

Equations (4) and (5), again, were based on morphometric data collected by Rombough and Moroz (1990).

Results

Mass-specific rates for total O_2 consumption estimated by adding uptake rates in the anterior and posterior compartments of the respirometer were not significantly different from those estimated for individual fish of roughly the same size without the rubber dam (Wilcoxon signed-rank test; $P < 0.1$, $n = 14$).

The rate of O_2 consumption increased in a parabolic fashion throughout the experimental period (fig. 2). Mass exponents were 0.480, 1.325, and 0.887, respectively, for $\dot{M}_{O_2c}$, $\dot{M}_{O_2b}$, and $\dot{M}_{O_2t}$. Corresponding R^2 values were 88.2%, 88.2%, and 90.6%, with $n = 49$ in all three cases.

Cutaneous uptake accounted for up to 84% of total O_2 consumption in newly hatched alevins (fig. 3). The relative amount of O_2 taken up across the skin, however, declined rapidly during the alevin period. By the time the alevins were about halfway through yolk absorption (i.e., at $m = 281$ mg), $\dot{M}_{O_2c}$ accounted for only 50% of $\dot{M}_{O_2t}$. At the start of exogenous feeding, the cutaneous contribution had fallen to about 40% of total O_2 consumption.

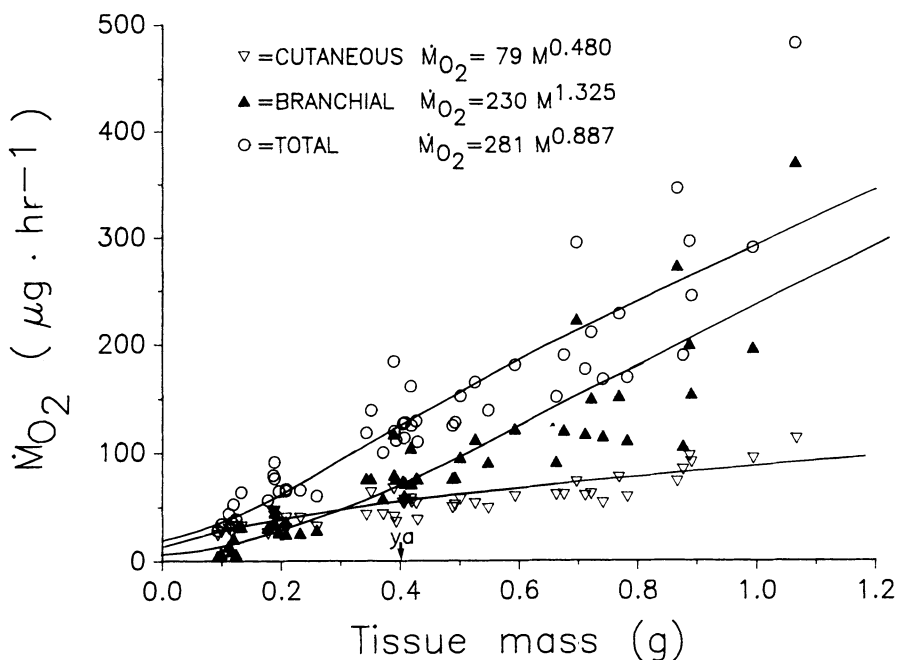


Fig. 2. Cutaneous (open triangles), branchial (closed triangles), and total (circles) O_2 uptake by chinook alevins and fry as functions of m . Equations describe lines of best fit. ya , m at which yolk absorption is essentially complete.

The relative importance of the skin as a site of gas exchange continued to decline during the early fry stage but at a slower rate. In a 1-g fish, about 30% of the O₂ consumed was taken up across cutaneous surfaces.

The decrease in the relative importance of the skin, and the corresponding increase in the relative importance of the gills, was partially due to a large increase in the efficiency of branchial gas exchange. At hatch the efficiencies of cutaneous and branchial surfaces, expressed in terms of area-specific oxygen uptake, were roughly the same (about 7.5 and 10.1 $\mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$, respectively; fig. 4). Area-specific cutaneous uptake (E_c) increased only slightly during the experimental period (to 9.6 $\mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ at 1.0 g). The line of best fit for E_c was

$$E_c = 7.19 + 2.37 m \quad (R^2 = 17.6\%, n = 49, P < 0.01). \quad (6)$$

In contrast, area-specific branchial uptake (E_b) increased almost sixfold, to about 60 $\mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ for a 1.0-g fish. The hyperbolic function

$$E_b = 127 m (1.16 + m)^{-1} \quad (R^2 = 62.8\%, n = 46) \quad (7)$$

provided the best fit to the data for E_b .

Discussion

The skin is the major site of gas exchange in newly hatched alevins. This is not particularly surprising given that, at hatch, A_c is about 23 times A_b (Rombough and Moroz 1990). What is surprising is the speed with which the gills supplant the skin as the primary site of $\dot{M}_{O_2}$. By the time the alevins weigh 281 mg (about halfway between hatch and the start of exogenous feeding), $\dot{M}_{O_2b}$ supplies about half of the total O₂ consumed. Branchial surface area at this stage (i.e., at $m = 281$ mg), however, is only 28% of total surface (Rombough and Moroz 1990). Indeed, in chinook the surface area of the gills does not match that of the skin until the fish weigh about 2.5 g (Rombough and Moroz 1990).

Only about half of the increase in $\dot{M}_{O_2b}$ during the alevin stage is due to the expansion in A_b . The other half appears to be due to an increase in the efficiency of branchial gas exchange. This increase in efficiency, expressed here in terms of area-specific uptake (fig. 4), probably reflects the major morphological and functional changes taking place in the gill as the fish mature. Chinook do not possess functional gill lamellae at hatch (Rombough and Moroz 1990). Gill lamellae, however, develop rapidly during the period

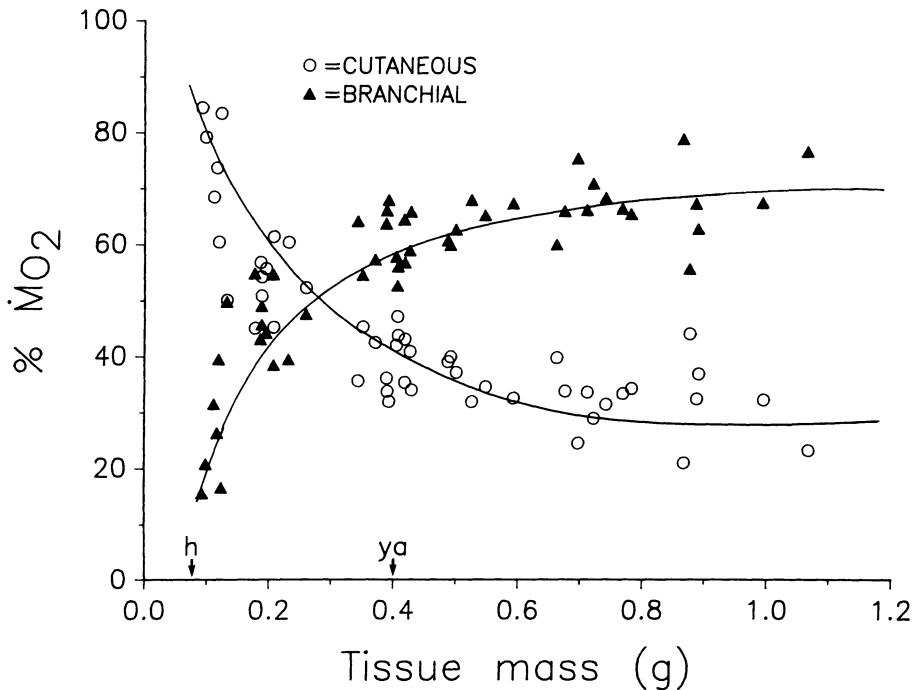


Fig. 3. Relative importance of cutaneous and branchial surfaces in O_2 uptake by chinook alevins and fry. Cutaneous (circles) and branchial (triangles) uptakes are expressed as percentages of total uptake. h, m at hatch; ya, m at complete yolk absorption.

shortly after hatch so that by 100 accumulated thermal units after hatch (mean temperature $\times$ number of days after hatch) the lamellae account for about half the total gill area (Rombough and Moroz 1990). Gas exchange is generally assumed to be more efficient across gill lamellae than across gill filaments (which account for virtually all of gill surface area at hatch). Lamellar blood flow is countercurrent to the ventilatory flow of water, while blood flow in the filaments is approximately at right angles to the flow. Although there are no empirical data, lamellar diffusion distances are probably less than filamental diffusion distances in alevins, just as they are in adults. Lamellar diffusion distances themselves gradually decrease as alevins mature (Morgan 1974). Ventilatory movements at hatch tend to be weak in salmonids, which suggests that the gills initially may be underventilated. Ventilatory movements become more forceful and better coordinated as alevins mature (Holeton 1971; Peterson 1975). Taken together, these changes are probably sufficient to account for the observed increase in the E_b of developing chinook.

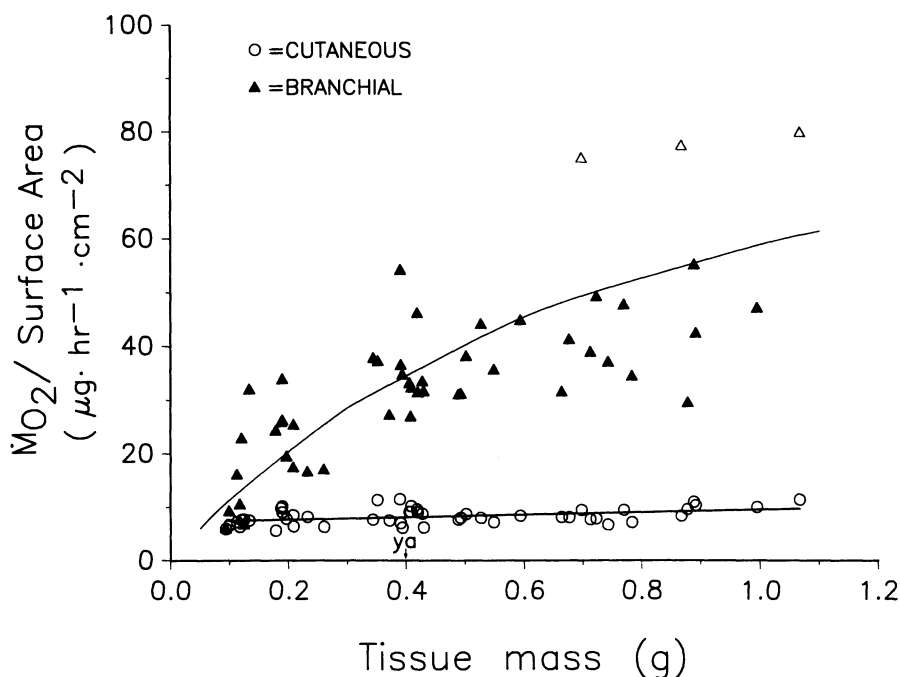


Fig. 4. Changes in area-specific uptakes across cutaneous (circles) and branchial (triangles) surfaces during the growth and development of chinook alevins and fry. Data points represented by the open triangles were excluded when the line of best fit for branchial uptake was calculated. ya, m at complete yolk absorption.

Since $\dot{M}O_2$ is highly dependent on factors such as temperature and activity, one must be careful in comparing the values for area-specific uptake obtained in these experiments with values reported in the literature for older fish. When this is kept in mind, the values we obtained for young chinook, particularly those for E_c agree surprisingly well with literature values for adult fish. Area-specific uptake across the skin averaged about $8.5 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ in young chinook. Area-specific uptakes of 9.9, 6.9, 8.2, and $6.5 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ were reported for the skin of adult rainbow trout *Salmo gairdneri*, eel *Anguilla anguilla*, tench *Tinca tinca*, (Kirsch and Nonnotte 1977) and plaice *Pleuronectes platessa* (Steffensen et al. 1981), respectively. Area-specific uptake across the gills increased hyperbolically from about 10 to $60 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ in young chinook. Data presented by Randall and Daxboeck (1984) for juvenile and adult fish indicate area-specific uptakes of about $19 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ for rainbow trout, $13\text{--}42 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ for catfish *Ictalurus nebulosus*, and $75 \mu\text{g} \cdot \text{h}^{-1} \cdot \text{cm}^{-2}$ for tuna *Thunnus albacares*.

Although the relative importance of the skin declined more rapidly than one might expect simply on the basis of its relative surface area, the skin remained a significant site of gas exchange well into the fry stage. Cutaneous uptake accounted for between 20% and 40% of total O_2 consumption in 19 of the 20 tests conducted with fish weighing between 0.45 and 1.1 g (fig. 3). Studies of cutaneous uptake in juveniles and adults of other species suggest that the relative importance of the skin in respiratory gas exchange declines relatively little as fish grow beyond this size. Values of 30%–35%, 27%, 22%, and 15% for cutaneous uptake as a fraction of total O_2 consumption have been reported for adults of eel (Kirsch and Nonnotte 1977; LeMoigne et al. 1986), plaice (Steffensen et al. 1981), tench (Kirsch and Nonnotte 1977), and rainbow trout (Kirsch and Nonnotte 1977), respectively. Indeed, extrapolation of the curve relating the relative importance of the skin in $\dot{M}O_2$ to tissue mass (fig. 3) suggests that cutaneous uptake in chinook will stabilize at about 30% of total O_2 consumption.

Acknowledgments

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Literature Cited

- DESILVA, C. 1974. Development of the respiratory system in herring and plaice larvae. Pages 465–485 in J. S. BLAXTER, ed. *The early life history of fish*. Springer, Berlin.
- HOLETON, G. F. 1971. Respiratory and circulatory responses of rainbow trout larvae to carbon monoxide and to hypoxia. *J. Exp. Biol.* 55:683–694.
- KIRSCH, R., and G. NONNOTTE. 1977. Cutaneous respiration in three freshwater teleosts. *Respir. Physiol.* 29:339–354.
- LEMOIGNE, J., P. SOULIER, M. PEYRAUD-WAITZENEGGER, and C. PEYRAUD. 1986. Cutaneous and gill O_2 uptake in the European eel (*Anguilla anguilla* L.) in relation to ambient PO_2 , 10–400 Torr. *Respir. Physiol.* 66:341–354.
- LIEM, K. F. 1981. Larvae of air-breathing fishes as countercurrent flow devices in hypoxic environments. *Science* 211:1177–1179.
- MORGAN, M. 1974. Development of secondary lamellae of the gills of the trout, *Salmo gairdneri* (Richardson). *Cell Tiss. Res.* 151:509–523.
- OIKAWA, S., and Y. ITAZAWA. 1985. Gill and body surface areas of the carp in relation to body mass, with special reference to the metabolism-size relationship. *J. Exp. Biol.* 117:1–14.

- PETERSON, R. H. 1975. Pectoral fin and opercular movements of Atlantic salmon (*Salmo salar*) alevins. J. Fisheries Res. Board Can. 32:643–648.
- RANDALL, D., and C. DAXBOECK. 1984. Oxygen and carbon dioxide transfer across fish gills. Pages 263–314 in W. S. HOAR and D. J. RANDALL, eds. Fish physiology. Vol. 10A. Academic Press, Orlando, Fla.
- ROMBOUGH, P. J. 1988a. Aerobic metabolism and dissolved oxygen requirements of embryos and alevins of chinook salmon (*Oncorhynchus tshawytscha*). Veroff. Univ. Innsbruck 167:95.
- . 1988b. Growth, aerobic metabolism and dissolved oxygen requirements of embryos and alevins of the steelhead trout, *Salmo gairdneri*. Can. J. Zool. 66:651–660.
- . 1988c. Respiratory gas exchange, aerobic metabolism, and effects of hypoxia during early life. Pages 59–161 in W. S. HOAR and D. J. RANDALL, eds. Fish physiology. Vol. 11A. Academic Press, Orlando, Fla.
- ROMBOUGH, P. J., and B. M. MOROZ. 1990. The scaling and potential importance of cutaneous and branchial surfaces in respiratory gas exchange in young chinook salmon (*Oncorhynchus tshawytscha*). J. Exp. Biol. 154:1–12.
- STEFFENSEN, J. F., and J. P. LOMHOLT. 1985. Cutaneous oxygen uptake and its relation to skin blood perfusion and ambient salinity in the plaice, *Pleuronectes platessa*. Comp. Biochem. Physiol. 81A:373–375.
- STEFFENSEN, J. F., J. P. LOMHOLT, and K. JOHANSEN. 1981. The relative importance of skin oxygen uptake in the naturally buried plaice, *Pleuronectes platessa*, exposed to graded hypoxia. Respir. Physiol. 44:269–272.
- TOULMOND, A., P. DEJOURS, and J. P. TRUCHOT. 1982. Cutaneous O₂ and CO₂ exchanges in the dogfish, *Scyliorhinus canicula*. Respir. Physiol. 48:169–181.